

POLLUTANTS AND THEIR EFFECTS ON ENVIRONMENT

Have you ever heard about acid rain? Do you know humans are contributor to climate change and heat waves we face every year? Our direct and indirect activities, like deforestation and building homes and industries impact on the environment and the species that live in the vicinity. These human activities add certain substances in the environment which damages it, these are termed

In this Chapter you will learn about:

- Air Pollutants (Sulphur Dioxide, Carbon Monoxide, Oxides of Nitrogen, Chlorofluorocarbons).
- Sources (Natural and From Human Activities).
- Harmful Effects (on Human Organ Systems: Lung Diseases, Brain Damage, Breathing, Headaches).
- Effects of Human Activity on Environment (Greenhouse Effect, Ozone Depletion and Global Warming, Acid Rain, Wild Life, Deforestation, Lack of Energy Resources).
- Saving the Earth (Solid Waste Management, Recycling of Materials, Conservation of Resources, Environmental Campaigns, Responsibility for All).

All the students will be able to

- Explain the sources, properties and harmful effects of air pollutants.
- List problems in human organ system caused by air pollutants.
- Plan and conduct a campaign that can help to reduce the air pollution in their local area.
- Explain the greenhouse effect.
- Describe the cause and effects of ozone depletion.
- Carry out a research to explain global warming and its effects on life on earth.
- Design a model to explain greenhouse effect.
- Explain the formation of acid rain and identify its consequences on living and non-living things.
- Define deforestation.
- State the effects of deforestation on the environment.
- Identify the human activities that have long term adverse effects on the environment.
- Explain the importance of local and global conservation of natural resources.
- Suggest ways in which individual, organization and government can help to make our earth a better place to live.



Fig.4.1: Air Pollution



Fig.4.2: Land Pollution



Fig.4.3: Deforestation

as pollutants. Let us explore different pollutants, their sources and effects on our environment, health and well-being.

POLLUTANTS, THEIR SOURCES AND HARMFUL EFFECTS ON HUMAN ORGAN SYSTEMS

- Explain the sources, properties and harmful effects of air pollutants.
- List problems in human organ system caused by air pollutants.
- Plan and conduct a campaign that can help to reduce the air pollution in their local area.

Conservation of quality of environment and its resources is the key to survival of our present and future generation. Unfortunately, **human being's** activities on earth are continuously altering the environment and making the survival of human race more difficult on this planet. Today **human being** is facing one of the most horrible ecological crises of his time i.e. pollution. Pollution is defined as the undesirable changes in the physical, chemical or biological characteristics of air, land and water that harmfully affect human life and other organisms.

Many substances can damage environment by making it dirty and unhealthy for the organisms living in the environment. The harmful substances which damage the environment are called pollutants and their introduction into the environment causes pollution. Air pollution is one of the most dangerous and common kind of environmental pollution that is being reported in most industrial cities of the World. It is caused when amount of solid wastes or concentration of gasses (other than Oxygen) increase in the air due to some natural processes or human activities. Some of the major air pollutants and their sources are mentioned below.

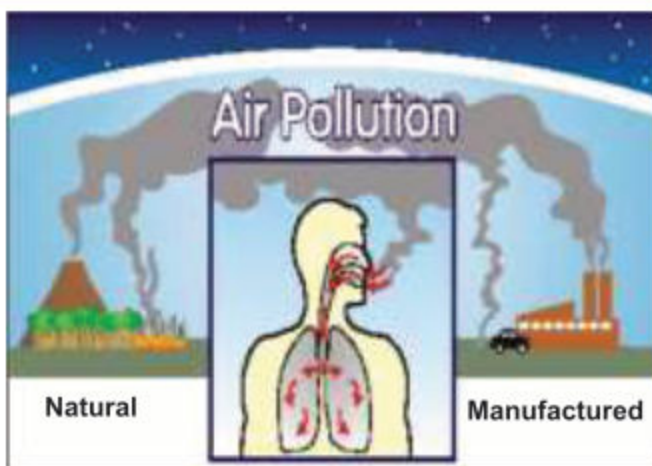


Fig.4.4: Sources of Air Pollution

1. Sulfur Dioxide

Sulphur Dioxide, a poisonous gas is one of the main by-products of industrial and motor vehicle emission having a terribly irritating smell. Constant exposure to Sulphur dioxide can lead to a higher occurrence of cough, sneeze, and cold, shortness of breath, bronchitis, and fatigue. Air infected with Sulphur Dioxide is the main reason for asthmatic attacks. Eye irritation, eye watering, shortness of breath, lung damage are other common problems found in individuals.



Fig.4.5: Effects of Air Pollution

2. Carbon Monoxide

Incomplete combustion of fuel leads to exhaust of Carbon Monoxide. This colourless and odorless gas mostly results in disease related to heart like heart attacks, cardiovascular diseases among others as well as headache and reduced mental awareness. Carbon Monoxide in excess may affect the lungs and lead to weakening of lung functions. It may have drastic effect on cardiovascular organs and could lead to fatigue and low productivity in a healthy individual.

3. Oxides of Nitrogen

Nitrogen dioxide is one pollutant mostly found at major road junctions and in heavy industrial areas. It is also one of the main contributors of smog and has a adverse effect on the human health. Too much exposure to Nitrogen dioxide can lead to various respiratory and lung diseases, infections, lung irritation and respiratory symptoms for example, cough, chest pain, difficulty in breathing.

4. Chlorofluorocarbon

Chlorofluorocarbons (CFC), a group of gasses which are widely used as refrigerants and industrial foaming agents. CFC is a major

cause of the depletion of ozone layer. Inhalation of high levels of chlorofluorocarbons can affect the lungs, central nervous system, heart, liver and kidneys. Symptoms of exposure to chlorofluorocarbons can include drowsiness, unclear speech, disorientation, tingling sensations and weakness in the limbs. Exposure to extremely high levels of chlorofluorocarbons can result in death. Ingestion of chlorofluorocarbons can also lead to nausea, irritation of the digestive tract and diarrhea.

Activity 4.1:

Plan and Conduct a Campaign to Reduce Air Pollution in the Local Area

Design a flyer or chart paper with all information regarding the source and impact of major air pollutants on the human organ systems. Suggest some ways to reduce the air pollution in local area. Give presentations to students of other classes in your school. Also, share this information with your family, neighbours, friends and relatives.

- Identify the human activities that have long adverse effects on the environment.

EFFECTS OF HUMAN ACTIVITIES ON ENVIRONMENT

A balanced relationship between living organisms, including humans and environment is essential to maintain life on earth. During last 100 years increased human population and wide spread industrialization societies have created a massive impact of human activities on the environment on earth. The unbalanced use of natural resources has created harmful impact on environment. For example, the change in our mode of transportation and heavy industrial wastes result in the emission of carbon dioxide, and other industrial gasses responsible for

Teacher Note:

Teacher should facilitate students in preparing information material within their groups and should also plan how to disseminate this information with other members in the local area.

global warming which could lead to flood. The fungicides, insecticides used on crops ultimately reach in soil water and hence affect the soils environment and plant growth. These practices produce harmful effects on the environment as mentioned below;

- Explain the formation of acid rain and identify its consequences on living and non-living.

1. Acid Rain

Power stations and other industrial units run by burning of coal processes emit carbon dioxide, carbon monoxide, sulfur dioxide, and Nitrogen oxide into the air which can dissolve in rain water to form acid rain. Acid rain affects the photosynthesis and growth of plants. When acid rain falls in rivers and lakes it can kill the aquatic organisms. Acid rain is not only a health hazard, but also can destroy many manmade structures, metals and statues of archaeological importance, kill many animals, destroy the agricultural land and cause damage to crop and plants.



Fig.4.6: Effects of Acid Rain

- Explain the greenhouse effect.
- Carry out a research to explain global warming and its effects on life on earth.
- Design a model to explain greenhouse effect.

2.Greenhouse Effect or Global warming

Carbon dioxide gas is produced when trees and fossil fuels are burnt. Carbon dioxide is a major greenhouse gas as it trap the sun heat and prevents it from escaping into space, just like the way a greenhouse traps heat from the sun. As more trees are burnt more Carbon dioxide is released into the atmosphere



Fig.4.7: The Greenhouse Effect

and greenhouse effect increases. As a result, the average temperature of the surface of the earth increases. This is known as global warming. Other gasses which contribute to the greenhouse effect are also being released in the atmosphere by human activities. These include methane, nitrous oxide and CFC (Chlorofluorocarbon).

Activity 4.2:

Design a Model on Green House Effect

Things we need: Two glasses, cold water, ice cubes, plastic bag, thermometer.

Steps: Fill two glasses of equal size each with 2 cups of cold water. Put five ice cubes in each glass, then wrap one in a plastic bag and seal it tightly. Place both glasses in the sun for one hour, then measure the water temperature in each glass using a household thermometer. What did you observe? The glass covered in plastic is warmer because the bag had trapped heat in the glass, in the same way the greenhouse gases trap heat in the atmosphere.

➤ Describe the cause and effects of ozone depletion.

3. Depletion of Ozone Layer

High up around the earth in its atmosphere, in between twelve to fifty kilometers above the ground, there is a layer of ozone gas (O_3). The Ozone protects the earth and the organisms from harmful effects of ultra violet rays of the sun. The release of chlorine atoms into the atmosphere break down the ozone shield. The major source of chlorine is Chlorofluorocarbon (CFC).

Due to the depletion of ozone layer, ultra violet light penetrates and reaches at the earth surface and causes skin cancer and many other lethal effects on many organisms including humans.



Fig.4.8: Ozone layer surrounding the Earth

- Define deforestation.
- State the effects of deforestation on the environment.

4. Deforestation

Forests are vital to our Earth. Trees purify our air, filter our water, prevent erosion, and act as a buffer against climate change. They offer a home to plant and animal species. While trees provide natural resources such as medicine, food, timber, and fuel. Deforestation can be defined as the permanent destruction of forests to make the land available for other uses. Major causes of deforestation include overpopulation, paper, mining, logging, agriculture expansion and climate change. Some of the effects of deforestation on environment and human life are mentioned below;



Fig.4.9: Deforestation

1. Increased Greenhouse Gas Emissions.

Trees help to mitigate carbon dioxide and other greenhouse gas emissions, but they become carbon sources once they're cut, burned, or otherwise removed.

2. Acidic Oceans.

The oceans are becoming more acidic with an increased supply of carbon dioxide from deforestation and burning fossil fuels, putting ocean species and ecosystems at an extreme risk.

3. Loss of Species.

Giant pandas, rhinos, and the Asian elephant are just a few of hundreds of endangered species due to deforestation.

4. Climate Control.

Trees block sun rays during the day and holds in the heat at night. Removing trees from the forest leads to extreme temperature swings that are harmful to plants and animals.

5. Flooding and Erosion.

Without trees, erosion often occurs and sweeps land into

nearby rivers. Erosion also results in the contamination of water supply which greatly decrease the quality of our drinking water.

6. Life Quality Decrease.

Millions of people in the world depend on forests for hunting, small-scale agriculture and medicine. Common materials we use every day such as latex, cork, fruit, nuts, natural oils, and resins are found in the tropical forests. Deforestation disrupts the lives of millions of people and leads to migration.

- Explain the importance of local and global conservation of natural resources.
- Suggest ways in which individual, organization and government can help to make our earth a better place to live.

SAVING THE EARTH

There is something called a balance in nature. As we continue to overuse natural resources, a serious imbalance has been caused. It is very important to conserve the natural resources which are essential for survival for example trees, water, energy. Natural resources conservation can be done in several ways. Simply making people aware and setting an example for others to follow are the biggest steps at individual level. Further, we can reduce pollution and save our natural resources by taking following actions at individual, organization and government level.

1. Plantation, forestation and protecting the wild life by banning on the products made from the skin and body of animals.
2. Use reusable and recycled material for example paper bags in place of plastic bags.
3. Avoid the unnecessary use of energy, for example switching off light and fans while leaving the room. Using stair instead of lifts.
4. Use technology to develop materials, products and processes which are environment friendly.
5. Use filters or scrubbers on industrial chimneys to remove sulfur dioxide.
6. Less use of fertilizers and pesticides.
7. Reduce water wastage in our daily activities and reuse water wherever possible.

SUMMARY

- Undesirable changes in the physical, chemical or biological characteristics of air water and land is called Pollution.
- The harmful substances which damage the environment are called pollutants.
- Acid rain, greenhouse effect or global warming, depletion of ozone layers are the consequences of pollution.
- We can conserve the natural resources by adopting good practices and policies at individual, organization and government level.

EXERCISE

Answer the following questions:

1. What are major air pollutants, their causes and effects on human body?
2. Define pollution.
3. What is the role of ozone layer in conserving the environment?
4. What is greenhouse effect?
5. Write down different ways to reduce pollution.

Choose the best answers:

1. The most dangerous type of pollution is:
a) Water pollution. b) air pollution.
c) Noise pollution. d) Land pollution.
2. Name the major cause of depletion of ozone layer:
a) Carbon dioxide. b) Sulphur dioxide.
c) Chlorofluorocarbon. d) Oxygen.
3. Which gas is emitted by incomplete combustion of fuel.
a) Carbon monoxide. b) chlorofluorocarbon.
c) Carbon dioxide. d) Sulphur dioxide.
4. The layer which protects the earth is called:
a) Ozone. b) oxygen layer.
c) CFC layer. d) Green house.
5. Which gas from the following is mainly responsible for Green House effect in the environment?
a) Nitrogen dioxide. b) Methane.
c) Sulphur Dioxide. d) Carbon Dioxide.
6. The phenomenon which leads to destruction of manmade structure and metal of archaeological importance is:
a) Floods. b) Tsunami.
c) Acid rain. d) Green House.

7. The step which can save the earth and natural resources:
- a) Wastage of water. b) gas engines in the air.
c) use recycled and reusable material. d) deforestation.
8. The animal included in the endangered species is:
- a) Deers. b) Giant pandas.
c) Jackals. d) Cat.
9. The poisonous gas which is the main byproduct of motor vehicle with irritating smell is:
- a) Oxygen. b) Sulphur Dioxide.
c) Silver Nitrate. d) Ammonia.
10. The ozone layer is present in the atmosphere approximately:
- a) 5 to 10 Km high up the ground.
b) 16 to 20 Km high up the ground.
c) 12 to 50 Km high up the ground.
d) 16 to 50 Km high up the ground.

Match the column (A) and (B)

Column(A)	Column (B)
Protection of earth.	Deforestation.
Permanent destruction of trees.	Save the earth.
Less use of fertilizers.	Exposure to CFC.
Drowsiness and unclear speech.	Pollution.
Ecological crisis.	Ozone Layer.